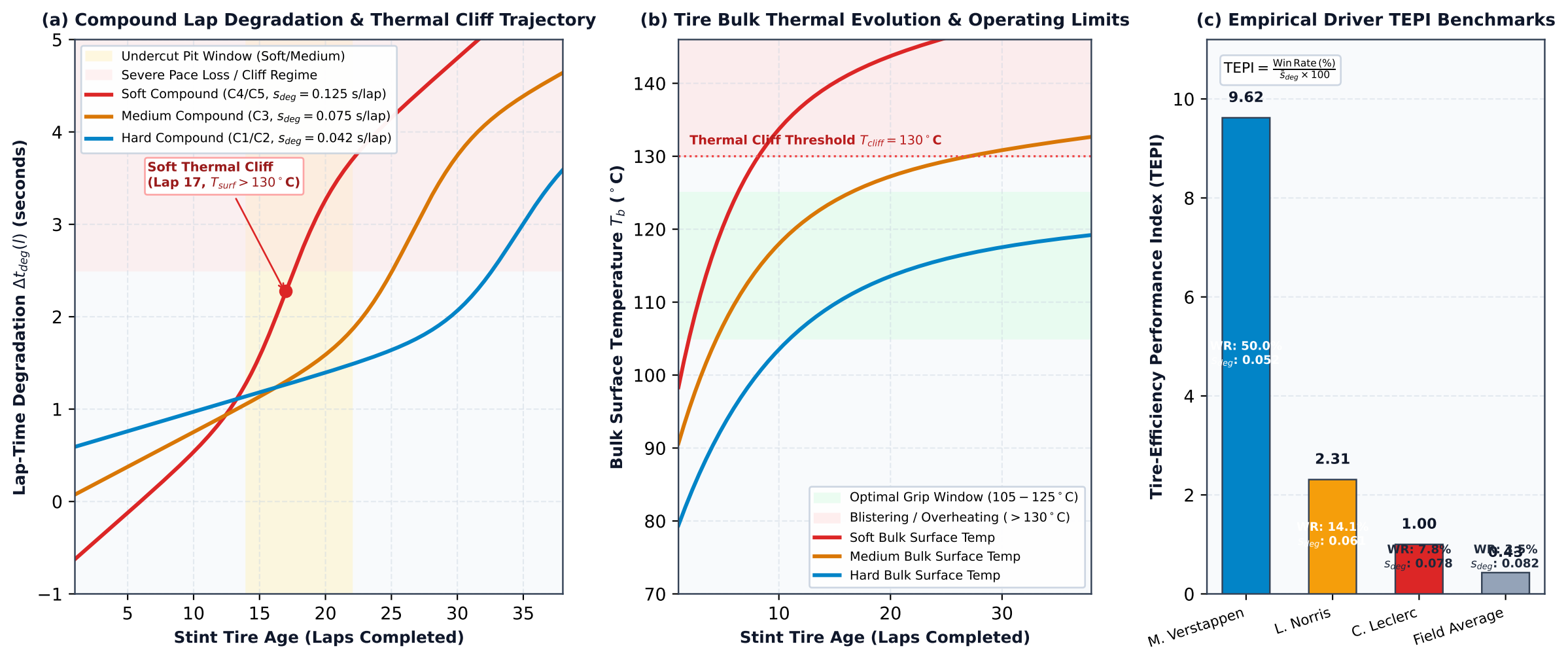

(a) Compound Lap Degradation & Thermal Cliff Trajectory
Undercut Pit Window (Soft/Medium)
Severe Pace Loss / Cliff Regime
Soft Compound (C4/C5, $s_{deg} = 0.125$ s/lap)
Medium Compound (C3, $s_{deg} = 0.075$ s/lap)
Hard Compound (C1/C2, $s_{deg} = 0.042$ s/lap)
Soft Thermal Cliff
(Lap 17, $T_{surf} > 130\,^\circ$C)
Lap-Time Degradation $\Delta t_{deg}(l)$ (seconds)
Stint Tire Age (Laps Completed)
(b) Tire Bulk Thermal Evolution & Operating Limits
Thermal Cliff Threshold $T_{cliff} = 130\,^\circ$C
Optimal Grip Window ($105 - 125\,^\circ$C)
Blistering / Overheating ($> 130\,^\circ$C)
Soft Bulk Surface Temp
Medium Bulk Surface Temp
Hard Bulk Surface Temp
Bulk Surface Temperature T_b ($^\circ$C)
Stint Tire Age (Laps Completed)
(c) Empirical Driver TEPI Benchmarks
TEPI = $\frac{\text{Win Rate (\%)}}{s_{deg} \times 100}$
Tire-Efficiency Performance Index (TEPI)
9.62
2.31
1.00
0.43
M. Verstappen
L. Norris
C. Leclerc
Field Average